

System Reliability

Fault Trees &

Discrete Bayesian Networks

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CHAPTER 1

Preliminaries

In this document we describe briefly what we understand by system reliability, how we can model system reliability through fault trees and how we can generalize the concept of a fault tree with the help of discrete Bayesian Networks. We start by remembering some basic concepts that might help for our purpose.

1.1 Axiomatic presentation of probability

Andrey Nikolaevich Kolmogorov (1903 - 1987) is recognized as the first to formalize probability theory as an axiomatic discipline similar to algebra, geometry and other branches of mathematics. An axiom is roughly a proposition that is self-evident and is accepted without a proof. In geometry for example, the concept of a “point” and a “line” are accepted as self-evident and are used in order to prove other more elaborate propositions. A formal derivation of probability theory as a mathematical discipline is out of the scope of these notes, however the main results concerning this derivation are presented. The main axioms of probability theory following Kolmogorov [1956] are presented next.

Let Ω be a collection of elementary events ξ, ζ, η and $\mathcal{F}$ a set of subsets of Ω ; the elements of the set $\mathcal{F}$ will be called random events.

Axiom I $\mathcal{F}$ is a field of sets. A system of sets is called a field if the sum, product and difference of two sets of the system also belong to the same system. Every non-empty field contains the empty set $\emptyset$.

Axiom II $\mathcal{F}$ contains Ω

Axiom III To each set A in $\mathcal{F}$ a non-negative real number $P(A)$ is assigned; $P(A)$ is called the probability of event A

Axiom IV $P(\Omega) = 1$

Axiom V If A and B have no element in common, that is $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$

A system of sets $\mathcal{F}$ together with a definite assignment of numbers $P(A)$ satisfying *Axiom I* till *Axiom V* is called a *probability field*.

The first and second axioms will not be discussed. *Axiom III* indicates that the probability P for a random event A is greater than or equal to 0. *Axiom IV* states that the probability of Ω is equal to 1. According to *Axiom V*, the probability of the occurrence of (A or B or both) is equal to the sum of the probabilities of A and B separately, provided that A and B are mutually exclusive.

Strictly speaking, the given axioms, especially *Axiom V*, are only valid as long as we restrict ourselves to outcomes with a finite number of outcomes. The extension to more general axioms (see Kolmogorov [1956]) contains no viewpoints that are important for our purposes and hence are left out.

In the above, probability is introduced as a mathematical quantity. This says little about the interpretation of the notion of probability. This interpretation is essentially a philosophical problem. The main interpretations of probability are briefly introduced below.

1.2 Interpretations of probability

Four of the main interpretations of probability are the classical, the logical, the frequentist and the subjective.

1. The *classical* interpretation of probability is often attributed to the French scientist Pierre-Simon Laplace (1749 - 1827). Laplace et al. [1951], defined probability as, “the number of favorable cases divided by the number of equi-possible cases”. Examples from coin tossing and dice-throwing are often used to illustrate what is meant by “equi-possible”. This interpretation however has been criticized regarding the operational definition of equi-possible. For example, for insurance companies death cases may not be equi-possible and hence has fallen into disuse.
2. The *logical* interpretation was proposed in Keynes [1921]. The basic idea underlying this interpretation was that conditional probability should be interpreted as partial entailment. This interpretation however is also in disuse and to a large extent forgotten.
3. The *frequentist* interpretation (Von Mises [1939]) introduces probability as limiting relative frequencies in a “collective” or “random sequence”. Where, roughly, a random sequence is one which passes all “recursive statistical tests”. For example the frequency of 1’s in a very large random sequence of 0’s and 1’s.
4. Finally the *subjective* interpretation. One of the main proponents is Savage [1954] and the interpretation of probability is in terms of degree of belief of a subject. Different subjects can have different degrees of belief for one and the same event. Again, very loosely, the theory would state that if a subject prefers A to B , for rational subjects this would entail that he or she assigns greater probability to A than to B .

1.3 The sample space

The concept of a sample space is very useful when we speak of outcomes of experiments or observations and their respective probabilities. This concept works especially well when we deal with a finite number of outcomes. In order to make this concept somehow more explicit (unfortunately not necessarily more formal), we follow the presentation of Feller [1968] who considers outcomes of experiments (for example simple ones such as coin tossing or throwing dice) to be sufficiently self-evident as to be used as the building blocks of more complex concepts. “Following a general usage in mathematics the simple events will be called *sample points*, or *points* for short. By definition, every indecomposable result of the (idealized) experiment is represented by one, and only one, sample point. The aggregate of all sample points will be called the *sample space*. All events connected with a given (idealized) experiment can be described as aggregates of sample points” [Feller, 1968, p.9].

Two examples provided by Feller [1968] are presented next.

Example 1.3.1. Table 1.1 presents the possible outcomes of the “experiment” of placing three balls into three cells. There are 27 sample points. The event that a appears in the first cell consists of 9 sample points.

Example 1.3.2. Table 1.2 presents the possible outcomes of the experiment of placing three indistinguishable balls into three cells. There are 10 sample points. The event that *three balls are placed in a single cell* consists of 3 sample points.

1.	{	abc		-		-		}	10.	{	a		bc		-		}	19.	{	-		a		bc		}
2.	{	-		abc		-		}	11.	{	b		ac		-		}	20.	{	-		b		ac		}
3.	{	-		-		abc		}	12.	{	c		ab		-		}	21.	{	-		c		ab		}
4.	{	ab		c		-		}	13.	{	a		-		bc		}	22.	{	a		b		c		}
5.	{	ac		b		-		}	14.	{	b		-		ac		}	23.	{	a		c		b		}
6.	{	bc		c		-		}	15.	{	c		-		ab		}	24.	{	b		a		c		}
7.	{	ab		-		c		}	16.	{	-		ab		c		}	25.	{	b		c		a		}
8.	{	ac		-		b		}	17.	{	-		ac		b		}	26.	{	c		a		b		}
9.	{	bc		-		a		}	18.	{	-		bc		a		}	27.	{	c		b		a		}

Table 1.1: Possible outcomes of placing three balls into three cells (Feller [1968])

1.	{	***		-		-		}	6.	{	*		**		-		}
2.	{	-		***		-		}	7.	{	*		-		**		}
3.	{	-		-		***		}	8.	{	-		**		*		}
4.	{	**		*		-		}	9.	{	-		*		**		}
5.	{	**		-		*		}	10.	{	*		*		*		}

Table 1.2: Possible outcomes of placing three indistinguishable balls into three cells (Feller [1968])

1.4 Basic probability calculus

1.4.1 Probability calculus with two events

Starting with the axioms from section 1.1, some propositions can be derived. These propositions may seem trivial. However, these proofs, on the basis of the Venn diagrams from Figures 1.1 to 1.10 illustrate the reasoning used in the calculus of probabilities.

Proposition 1.4.1. $P(\emptyset) = 0$

Proof. Because $P(A \cap \emptyset) = P(\emptyset)$ from *Axiom V* we have $P(A \cup \emptyset) = P(A) + P(\emptyset)$. Furthermore $(A \cup \emptyset) = A$ results in $P(A \cup \emptyset) = P(A)$. Combining these results gives $P(\emptyset) = 0$. $\square$

Proposition 1.4.2. $P(A) + P(\bar{A}) = 1$, where $\bar{A}$ is the complement of A or not A

Proof. Due to $A \cup \bar{A} = \Omega$ and $A \cap \bar{A} = \emptyset$ (see Figure 1.5) it follows according to *Axiom IV* and *Axiom V* that $P(A) + P(\bar{A}) = P(A \cup \bar{A}) = P(\Omega) = 1$. $\square$

Proposition 1.4.3. $0 \leq P(A) \leq 1$

Proof. The left inequality follows directly from *Axiom III*. Again, due to *Axiom III* it follows that $P(\bar{A}) \geq 0$. The right inequality follows from Proposition 1.4.2. $\square$

Proposition 1.4.4. If $A \subset B$ then $P(A) \leq P(B)$

Proof. Since $A \subset B$ is $(B - A) \cup A = B$ and $(B - A) \cap A = \emptyset$ (see Figure 1.6). Applying *Axiom V*: $P(B) = P((B - A) \cup A) = P(B - A) + P(A)$. With $P(B - A) \geq 0$ (*Axiom III*) it follows that $P(B) \geq P(A)$. $\square$

Proposition 1.4.5. If $A \subset B$ then $P(A \cup B) = P(B)$

Proof. $A \cup B = A \cup (B - A)$ is valid (see Figure 1.6), from which the proposition follows directly. $\square$

Proposition 1.4.6. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Proof. $A \cup B = A \cup (B - A)$ is valid (see Figure 1.7), as long as A and $(B - A)$ are mutually exclusive. Due to *Axiom V*: $P(A \cup B) = P(A) + P(B - A)$. Then $B = (A \cap B) \cup (B - A)$ is valid (see Figure 1.8), with $(A \cap B)$ and $(B - A)$ also being mutually exclusive. Due to *Axiom V*: $P(B) = P(A \cap B) + P(B - A)$. The substitution of $P(B - A)$ proves this proposition. $\square$

Proposition 1.4.7. $\max\{P(A), P(B)\} \leq P(A \cup B) \leq P(A) + P(B)$

Proof. For the left inequality $P(A \cup B) = P(A) + P(B - A)$ is used (see the proof of proposition 1.4.6 first step). Since $P(B - A) \geq 0$ (*Axiom III*) it holds that $P(A \cup B) \geq P(A)$. In exactly the same way one may show that $P(A \cup B) \geq P(B)$. The right inequality follows from proposition (1.4.6) and the fact that $P(A \cap B) \geq 0$ (from *Axiom III*). $\square$

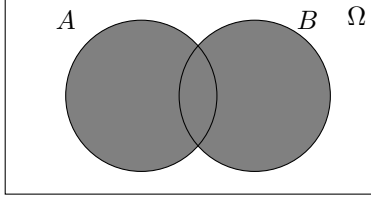


Figure 1.1: Union of events A and B : $A \cup B$; A or B

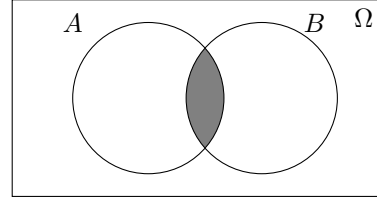


Figure 1.2: Intersection of events A and B : $A \cap B$; A and B

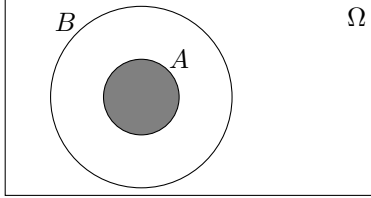


Figure 1.3: A is a subset of B : $A \subset B$; A is a part of B or B contains A

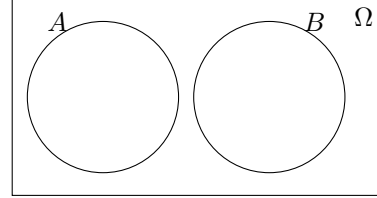


Figure 1.4: A and B are mutually exclusive $A \cap B = \emptyset$

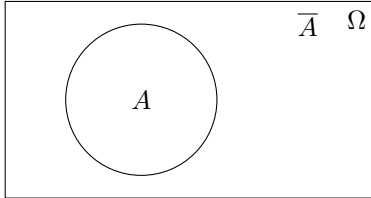


Figure 1.5: $A \cup \bar{A} = \Omega$ and $A \cap \bar{A} = \emptyset$; $\bar{A}$ is called the complement of A

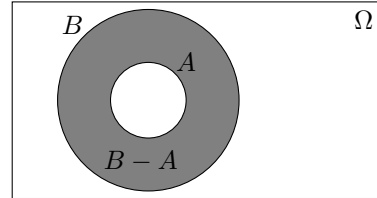


Figure 1.6: A is a subset of B then: $A \cup (B - A) = B$; $A \cap (B - A) = \emptyset$ and $A \cup B = B$

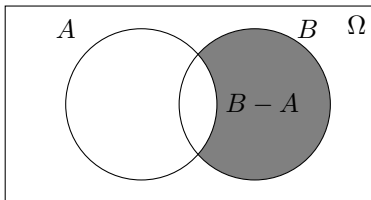


Figure 1.7: $A \cup (B - A) = A \cup B$, $A \cap (B - A) = \emptyset$

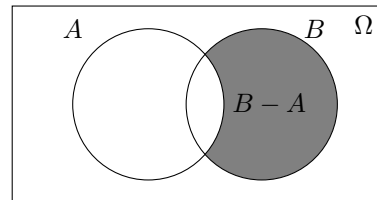


Figure 1.8: $(B - A) \cup (A \cap B) = B$, $(B - A) \cap (A \cap B) = \emptyset$

It is now interesting to relate the various propositions regarding $P(A \cup B)$. Proposition 1.4.6 gives an exact expression. Notice that in this case $P(A \cap B)$ has yet to be determined. Proposition 1.4.7 shows an upper and a lower bound expressed in terms of $P(A)$ and $P(B)$. The upper limit is exact

if A and B are mutually exclusive (see *Axiom V* and proposition 1.4.6); the lower limit is exact if $A \subset B$ or vice versa (see proposition 1.4.5).

Example 1.4.8. Consider a system that has two failure modes or consider only two failure modes of a particular system. Denote these failure modes A and B . Assume $P(A) = 2 \cdot 10^{-4}$ and $P(B) = 3 \cdot 10^{-4}$. For the event of failure of the system Proposition 1.4.7 states $\max\{P(A), P(B)\} \leq P(A \cup B) \leq P(A) + P(B)$, therefore:

$$3 \cdot 10^{-4} \leq P(F) \leq 5 \cdot 10^{-4} \quad (1.1)$$

Knowing these bounds may be sufficient for a number of problems. Note that the bounds can get closer together if one of the probabilities is much larger than the other one.

1.4.2 Probability calculus with more than two events

Some of the propositions can now be extended to multiple events. A proof is only given for proposition 1.4.9, all other propositions follow a similar reasoning.

Proposition 1.4.9. *If $A_i \cap A_j = \emptyset$ for every $i \neq j$ then: $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$*

Proof. This is an extension of *Axiom V*. For $n = 3$ the proposition is proven as follows: $P(A_1 + A_2 + A_3) = P((A_1 \cup A_2) \cup A_3) = P(A_1 \cup A_2) + P(A_3) = P(A_1) + P(A_2) + P(A_3)$. In this proof *Axiom V* is first applied to $(A_1 \cup A_2)$ and A_3 and then again to A_1 and A_2 . The extension to $n = 4, 5, \dots$ continues in the same manner and the proof is completed by induction. $\square$

Proposition 1.4.10. *If $A_i \cap A_j = \emptyset$ for every $i \neq j$ and $(A_1 \cup A_2 \cup \dots \cup A_n) = \Omega$ then: $P(A_1) + P(A_2) + \dots + P(A_n) = 1$*

This is an extension of proposition 1.4.2 and a special case of proposition 1.4.9.

Proposition 1.4.11. *If $A_i \subset A_j$ for every $i \neq j$ as an extension to proposition 1.4.5: $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_j)$.*

Proposition 1.4.12. *For an arbitrary A_i it holds that $\max\{P(A_i)\} \leq P(A_1 \cup A_2 \cup \dots \cup A_n) \leq \sum_i P(A_i)$.*

This is an extension of proposition 1.4.7. The probability that at least one of the events occurs is greater than the probability that the most probable event occurs and smaller than the sum of the probabilities of the separate events. The lower bound occurs when one of the events includes every other event (Figure 1.9). The upper bound occurs when all events are mutually exclusive (see Figure 1.10). A special case is the case when every probability $P(A_i)$ is equal:

$$P(A_i) \leq P(A_1 \cup A_2 \cup \dots \cup A_n) \leq n \cdot P(A_i) \quad (1.2)$$

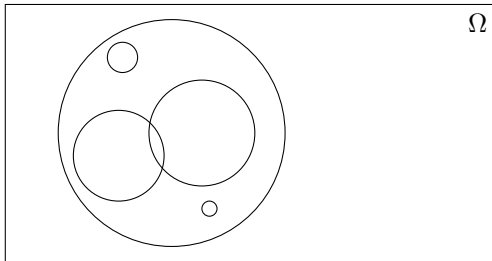


Figure 1.9: $P(A_1 \cup A_2 \cup \dots \cup A_n) = \max\{P(A_i)\}$

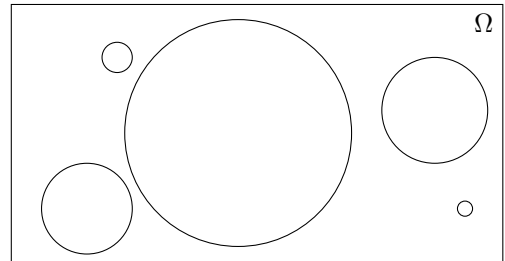


Figure 1.10: $P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_i P(A_i)$

Example 1.4.13. Take a six-sided fair die. Let A_i denote the event that a throw of the die yields the result i . The events A_i are mutually exclusive. Since the die is fair $P(A_i) = P(A_j)$, it follows on the basis of proposition 1.4.10 that $P(A_i) = 1/6$ for every i .

Example 1.4.14. The probability of throwing a 4, 5 or a 6 in one throw is, according to proposition 1.4.12: $1/6 \leq P(A_4 \cup A_5 \cup A_6) \leq 3/6$. Because the events A_4 , A_5 and A_6 are mutually exclusive, we know from proposition 1.4.9 that the right equality holds, that is $P(A_4 \cup A_5 \cup A_6) = 3/6$.

Example 1.4.15. Consider the outcomes of throwing two fair dice or one fair die twice. What is $P(\text{of throwing } 6 \cup 6)$? Table 1.3 shows all possible outcomes of throwing two dice. For example $\{1,1\}$ indicates that the outcome on the first and second throws was 1. It is easy to verify (by enumeration for example) that $P(\text{of throwing } 6 \cup 6) = 11/36$

1. $\{1,1\}$	7. $\{1,2\}$	13. $\{1,3\}$	19. $\{1,4\}$	25. $\{1,5\}$	31. $\{1,6\}$
2. $\{2,1\}$	8. $\{2,2\}$	14. $\{2,3\}$	20. $\{2,4\}$	26. $\{2,5\}$	32. $\{2,6\}$
3. $\{3,1\}$	9. $\{3,2\}$	15. $\{3,3\}$	21. $\{3,4\}$	27. $\{3,5\}$	33. $\{3,6\}$
4. $\{4,1\}$	10. $\{4,2\}$	16. $\{4,3\}$	22. $\{4,4\}$	28. $\{4,5\}$	34. $\{4,6\}$
5. $\{5,1\}$	11. $\{5,2\}$	17. $\{5,3\}$	23. $\{5,4\}$	29. $\{5,5\}$	35. $\{5,6\}$
6. $\{6,1\}$	12. $\{6,2\}$	18. $\{6,3\}$	24. $\{6,4\}$	30. $\{6,5\}$	36. $\{6,6\}$

Table 1.3: Possible outcomes of throwing 2 dice.

Example 1.4.16. Consider the outcomes of throwing three fair dice or one fair die three times. What is $P(\text{of throwing } 6 \cup 6 \cup 6)$? Table 1.4 shows all possible outcomes of throwing three dice. For example $\{1, 1, 1\}$ indicates that the outcome on the first, second and third throws was 1. The reader may still verify by enumeration that $P(\text{of throwing } 6 \cup 6 \cup 6) = 91/216$

Example 1.4.17. The probability of throwing a 6 in one throw is $1/6$. Because of proposition 1.4.12 the probability of throwing at least one 6 in three throws is greater than $1/6$ but smaller than $3/6$. None of these bounds is accurate. Explain why are these bounds not accurate.

Based on the examples 1.4.16 and 1.4.17 we can generalize proposition 1.4.6 to three dice as in equation 1.3. For a graphical representation using Venn diagrams see figure 1.11.

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3) \quad (1.3)$$

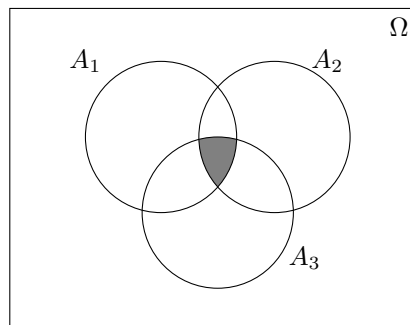


Figure 1.11: Union of three events: $A_1 \cup A_2 \cup A_3$

Example 1.4.18. A system will be built for a service life of 100 years. Each year the probability of failure is estimated at 10^{-5} . For the probability of failure in 100 years it applies: $10^{-5} \leq P(\text{failure in 100 years}) \leq 10^{-3}$ (see equation 1.2). Notice that there are two orders of magnitude between these bounds.

1. {1, 1, 1 }	37. {1, 1, 2 }	73. {1, 1, 3 }	109. {1, 1, 4 }	145. {1, 1, 5 }	181. {1, 1, 6 }
2. {1, 2, 1 }	38. {1, 2, 2 }	74. {1, 2, 3 }	110. {1, 2, 4 }	146. {1, 2, 5 }	182. {1, 2, 6 }
3. {1, 3, 1 }	39. {1, 3, 2 }	75. {1, 3, 3 }	111. {1, 3, 4 }	147. {1, 3, 5 }	183. {1, 3, 6 }
4. {1, 4, 1 }	40. {1, 4, 2 }	76. {1, 4, 3 }	112. {1, 4, 4 }	148. {1, 4, 5 }	184. {1, 4, 6 }
5. {1, 5, 1 }	41. {1, 5, 2 }	77. {1, 5, 3 }	113. {1, 5, 4 }	149. {1, 5, 5 }	185. {1, 5, 6 }
6. {1, 6, 1 }	42. {1, 6, 2 }	78. {1, 6, 3 }	114. {1, 6, 4 }	150. {1, 6, 5 }	186. {1, 6, 6 }
7. {2, 1, 1 }	43. {2, 1, 2 }	79. {2, 1, 3 }	115. {2, 1, 4 }	151. {2, 1, 5 }	187. {2, 1, 6 }
8. {2, 2, 1 }	44. {2, 2, 2 }	80. {2, 2, 3 }	116. {2, 2, 4 }	152. {2, 2, 5 }	188. {2, 2, 6 }
9. {2, 3, 1 }	45. {2, 3, 2 }	81. {2, 3, 3 }	117. {2, 3, 4 }	153. {2, 3, 5 }	189. {2, 3, 6 }
10. {2, 4, 1 }	46. {2, 4, 2 }	82. {2, 4, 3 }	118. {2, 4, 4 }	154. {2, 4, 5 }	190. {2, 4, 6 }
11. {2, 5, 1 }	47. {2, 5, 2 }	83. {2, 5, 3 }	119. {2, 5, 4 }	155. {2, 5, 5 }	191. {2, 5, 6 }
12. {2, 6, 1 }	48. {2, 6, 2 }	84. {2, 6, 3 }	120. {2, 6, 4 }	156. {2, 6, 5 }	192. {2, 6, 6 }
13. {3, 1, 1 }	49. {3, 1, 2 }	85. {3, 1, 3 }	121. {3, 1, 4 }	157. {3, 1, 5 }	193. {3, 1, 6 }
14. {3, 2, 1 }	50. {3, 2, 2 }	86. {3, 2, 3 }	122. {3, 2, 4 }	158. {3, 2, 5 }	194. {3, 2, 6 }
15. {3, 3, 1 }	51. {3, 3, 2 }	87. {3, 3, 3 }	123. {3, 3, 4 }	159. {3, 3, 5 }	195. {3, 3, 6 }
16. {3, 4, 1 }	52. {3, 4, 2 }	88. {3, 4, 3 }	124. {3, 4, 4 }	160. {3, 4, 5 }	196. {3, 4, 6 }
17. {3, 5, 1 }	53. {3, 5, 2 }	89. {3, 5, 3 }	125. {3, 5, 4 }	161. {3, 5, 5 }	197. {3, 5, 6 }
18. {3, 6, 1 }	54. {3, 6, 2 }	90. {3, 6, 3 }	126. {3, 6, 4 }	162. {3, 6, 5 }	198. {3, 6, 6 }
19. {4, 1, 1 }	55. {4, 1, 2 }	91. {4, 1, 3 }	127. {4, 1, 4 }	163. {4, 1, 5 }	199. {4, 1, 6 }
20. {4, 2, 1 }	56. {4, 2, 2 }	92. {4, 2, 3 }	128. {4, 2, 4 }	164. {4, 2, 5 }	200. {4, 2, 6 }
21. {4, 3, 1 }	57. {4, 3, 2 }	93. {4, 3, 3 }	129. {4, 3, 4 }	165. {4, 3, 5 }	201. {4, 3, 6 }
22. {4, 4, 1 }	58. {4, 4, 2 }	94. {4, 4, 3 }	130. {4, 4, 4 }	166. {4, 4, 5 }	202. {4, 4, 6 }
23. {4, 5, 1 }	59. {4, 5, 2 }	95. {4, 5, 3 }	131. {4, 5, 4 }	167. {4, 5, 5 }	203. {4, 5, 6 }
24. {4, 6, 1 }	60. {4, 6, 2 }	96. {4, 6, 3 }	132. {4, 6, 4 }	168. {4, 6, 5 }	204. {4, 6, 6 }
25. {5, 1, 1 }	61. {5, 1, 2 }	97. {5, 1, 3 }	133. {5, 1, 4 }	169. {5, 1, 5 }	205. {5, 1, 6 }
26. {5, 2, 1 }	62. {5, 2, 2 }	98. {5, 2, 3 }	134. {5, 2, 4 }	170. {5, 2, 5 }	206. {5, 2, 6 }
27. {5, 3, 1 }	63. {5, 3, 2 }	99. {5, 3, 3 }	135. {5, 3, 4 }	171. {5, 3, 5 }	207. {5, 3, 6 }
28. {5, 4, 1 }	64. {5, 4, 2 }	100. {5, 4, 3 }	136. {5, 4, 4 }	172. {5, 4, 5 }	208. {5, 4, 6 }
29. {5, 5, 1 }	65. {5, 5, 2 }	101. {5, 5, 3 }	137. {5, 5, 4 }	173. {5, 5, 5 }	209. {5, 5, 6 }
30. {5, 6, 1 }	66. {5, 6, 2 }	102. {5, 6, 3 }	138. {5, 6, 4 }	174. {5, 6, 5 }	210. {5, 6, 6 }
31. {6, 1, 1 }	67. {6, 1, 2 }	103. {6, 1, 3 }	139. {6, 1, 4 }	175. {6, 1, 5 }	211. {6, 1, 6 }
32. {6, 2, 1 }	68. {6, 2, 2 }	104. {6, 2, 3 }	140. {6, 2, 4 }	176. {6, 2, 5 }	212. {6, 2, 6 }
33. {6, 3, 1 }	69. {6, 3, 2 }	105. {6, 3, 3 }	141. {6, 3, 4 }	177. {6, 3, 5 }	213. {6, 3, 6 }
34. {6, 4, 1 }	70. {6, 4, 2 }	106. {6, 4, 3 }	142. {6, 4, 4 }	178. {6, 4, 5 }	214. {6, 4, 6 }
35. {6, 5, 1 }	71. {6, 5, 2 }	107. {6, 5, 3 }	143. {6, 5, 4 }	179. {6, 5, 5 }	215. {6, 5, 6 }
36. {6, 6, 1 }	72. {6, 6, 2 }	108. {6, 6, 3 }	144. {6, 6, 4 }	188. {6, 6, 5 }	216. {6, 6, 6 }

Table 1.4: Possible outcomes of throwing 3 dice.

1.4.3 Conditional probability

An important concept in probability theory is that of *conditional probability* for two events it is defined as follows:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (1.4)$$

Conditional probability (equation 1.4) is obviously only defined if $P(B) \neq 0$. Before we discuss the interpretation of conditional probability we first show it is most certainly a probability.

Proposition 1.4.19. *Axiom III, Axiom IV and Axiom V hold for the definition of conditional probability according to equation 1.4.*

Proof. Axiom III follows naturally. Because $P(A \cap B) > 0$ and $P(B) \neq 0$ it must follow that $P(A|B) > 0$.

Consider next *Axiom IV*. The proof that this axiom holds is as follows:

$$P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

In order to appreciate the statement above see Figure 1.3. Notice that from Figure 1.3 it holds that:

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

Finally it must be shown that for equation (1.4) *Axiom V* holds. We use the Venn-diagram from Figure 1.12. A_1 and A_2 are mutually exclusive and each one has an intersection with B .

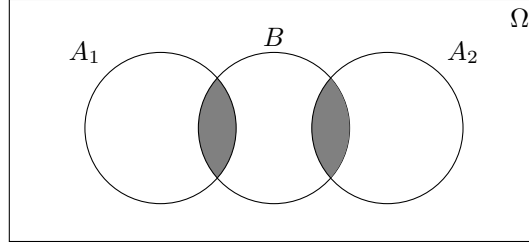


Figure 1.12: Intersection of events A and B : $A \cap B$; A and B

It can be concluded that:

$$\{(A_1 \cup A_2) \cap B\} = (A_1 \cap B) \cup (A_2 \cap B) \quad (1.5)$$

Since A_1 and A_2 are mutually exclusive, $(A_1 \cap B)$ and $(A_2 \cap B)$ are also mutually exclusive. This means:

$$P\{(A_1 \cup A_2) \cap B\} = P(A_1 \cap B) + P(A_2 \cap B) \quad (1.6)$$

If we divide the left part as well as the right part of the equation above by $P(B) \neq 0$:

$$\frac{P\{(A_1 \cup A_2) \cap B\}}{P(B)} = \frac{P(A_1 \cap B)}{P(B)} + \frac{P(A_2 \cap B)}{P(B)} \quad (1.7)$$

Then, using the definition of conditional probability (equation (1.4)) :

$$P\{(A_1 \cup A_2)|B\} = P(A_1|B) + P(A_2|B) \quad (1.8)$$

This last equation is *Axiom V* for conditional probabilities which concludes the proof. $\square$

Because the basic axioms of probability theory hold for conditional probability, all propositions for non-conditional probabilities can be rewritten for conditional probabilities. For example:

- $P(A|B) \leq 1$ which is analogue with *Axiom I* in section 1.1.
- $P(A|B) + P(\bar{A}|B) = 1$ as in *Axiom V*.
- $P(\emptyset|B) = 0$ similarly to *Axiom IV*.

In other words, it is possible to interpret the conditional probability as “usual” probability where the total sample space Ω is reduced to a particular set of events B . An important argument for this interpretation is:

$$P(B|B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1 \quad (1.9)$$

In this case, event B plays the role of Ω which has a probability of occurrence of 1. This brings us to the interpretation of B as a “given”. For some reason or another the event B can be considered to have occurred. Notice that the following holds:

- Let $A \cap B = \emptyset$, then $P(A|B) = 0$. This may be seen in Figure 1.4.
- Let $A \subset B$ then $P(A|B) = P(A \cap B)/P(B) = P(A|B)$. Notice that for more events, this procedure would normalize every probability of events $A_i \subset B$ without compromising their ratios (Figure 1.3).
- Let $(A \cap B) \neq \emptyset$, then A can be broken down in the separate events $(A \cap B)$ and $\bar{A} \cap B$. This leads to equation (1.4). The resulting probability $P(A|B)$ is usually called the probability of event A given event B (Figure 1.2).

1.4.4 Dependence and Independence

In general $P(A|B)$ will not be the same as $P(A)$. In the special case that this holds we speak of *independence*. Two events A and B are independent (and we denote independence as $\perp$) if:

$$P(A|B) = P(A) \quad (1.10)$$

Intuitively, equation 1.10 says that A is independent of B if the fact that B has occurred does not influence the probability of occurrence of A . Or in other words, knowing B for sure does not have an influence on your uncertainty over A . We are now in a position to say something about the rules for the probabilities of events defined by logical AND and OR operators in the case of independence (see proposition 1.4.6).

$$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B) \quad (1.11)$$

Notice that equation 1.11 implies that $P(A \cap B) = P(A) \cdot P(B)$. Mathematically it is better to use $P(A \cap B) = P(A) \cdot P(B)$ as a definition by itself instead of a result of independence. This definition may then be easily extended to n number of events. The events A_i are independent if:

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2) \cdot \dots \cdot P(A_n) \quad (1.12)$$

Calculating with independent events is mathematically trivial though could be computationally expensive. See for example equation 1.3. On the other hand, the adequate assignment of probabilistic dependence usually proves to be a challenging task. Therefore, there is a tendency to work under the assumption of independence. In practice, independence, or rather dependence, is based on physical attributes of the events (or variables) under consideration. Assuming independence when in fact the contrary is true may lead to large inaccuracies. Modelling of dependence is thus becoming increasingly important. Decision makers have realized the importance of more accurate probabilistic models and the traditional tools of probability and statistics need to be extended with models for dependence if we want to meet the challenge placed by the questions of interest posed by decision makers.

1.4.4.1 Independent dice

We have already discussed examples of independent events with the aid of throwing dice. The calculus of probabilities has been done mostly through enumeration. In this section we discuss further some examples where the calculus of probabilities is done mostly using independence. The reader may verify the results also by enumeration for example with the aid of tables 1.3 and 1.4.

Example 1.4.20. Denote by $A_1, \dots, A_6$ the event of showing a $1, \dots, 6$ in a throw of a fair die. Denote by E the event that the throw results in an even number. Then we have the following:

$$\begin{aligned} P(A_1|E) &= P(A_3|E) = P(A_5|E) = 0 \\ P(A_2|E) &= P(A_4|E) = P(A_6|E) = 1/3 \end{aligned}$$

Notice that, for example E and A_6 are not independent because $P(A_6|E) \neq P(A_6)$.

Example 1.4.21. Following with the notation from Example 1.4.20, one may verify that the events $E = \text{“throw is even”}$ and $A_{56} = \text{“throw is a 5 or 6”}$ are independent because:

$$\begin{aligned} P(E) &= P(A_2 \cup A_4 \cup A_6) = 1/2 \\ P(A_{56}) &= P(A_5 \cup A_6) = 1/3 \\ P(E \cap A_{56}) &= P(A_6) = 1/6 \end{aligned}$$

So it holds that $P(E \cap A_{56}) = P(E) \cdot P(A_{56})$

Example 1.4.22. In Example 1.4.16 a fair die was thrown three times and required was the probability that at least one of those throws is a 6. Now, the assumption is made that those results are independent and the probability can be calculated. Notice that the assumption of independence results from the fact that the die is fair. It is not true that probability theory indicates that these events should be independent. Denote:

$$\begin{aligned} B_1 &= \text{first throw is a 6} \\ B_2 &= \text{second throw is a 6} \\ B_3 &= \text{third throw is a 6} \\ B &= B_1 \cup B_2 \cup B_3 \end{aligned}$$

The example can be solved most easily by stating the problem slightly different: What is the probability that none of the throws result in a 6? This can be described as:

$$\overline{B} = \overline{B_1} \cap \overline{B_2} \cap \overline{B_3}$$

Because of the independence of $\overline{B_1}$ it holds (see (1.12)):

$$\begin{aligned} P(\overline{B}) &= P(\overline{B_1}) \cdot P(\overline{B_2}) \cdot P(\overline{B_3}) = P(\overline{B_i})^3 \\ 1 - P(B) &= \{1 - P(B_i)\}^3 \\ P(B) &= 1 - \{1 - P(B_i)\}^3 \\ P(B) &= \{1 - 1/6\}^3 \approx 0.42 \end{aligned}$$

See Example 1.4.17 for the upper and lower bound. Notice that equation 1.3 provides an alternative calculation.

1.4.4.2 Dependent dice

Von Mises [1939] discussed the case of dependent dice. He described the situation of two dice joined together by an invisible thread that would make particular outcomes in the throw of two dice more probable than others. In present times, it is not difficult to imagine this situation. Non-fair dice with certain magnetic properties could relatively easily be produced that would make the throw of two dice not independent.

Example 1.4.23. The outcome of throwing a die is denoted A_i for $i = 1, \dots, 6$ according to the outcome observed. Similarly, the outcome of throwing a second die is denoted B_i . Consider $P(A_i) = P(B_i) = 1/6$ and:

$$\begin{aligned} P(B_i \text{ is odd} | A_i \text{ is odd}) &= P(B_i \text{ is even} | A_i \text{ is even}) = 17/54 \\ P(B_i \text{ is even} | A_i \text{ is odd}) &= P(B_i \text{ is odd} | A_i \text{ is even}) = 1/54 \end{aligned}$$

It is easy to verify that:

$$\begin{aligned} P(B_i \text{ is odd} \cap A_i \text{ is odd}) &= P(B_i \text{ is even} \cap A_i \text{ is even}) = 17/324 \\ P(B_i \text{ is even} \cap A_i \text{ is odd}) &= P(B_i \text{ is odd} \cap A_i \text{ is even}) = 1/324 \end{aligned}$$

A graphical representation of these probabilities is given in Figure 1.13.

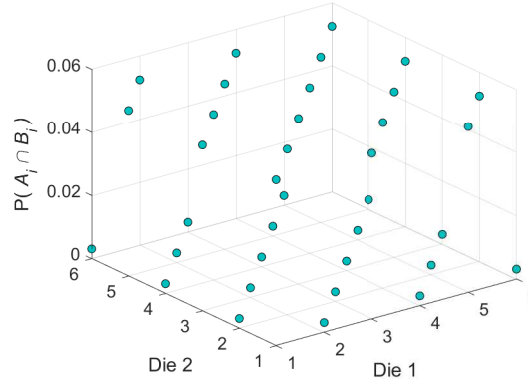


Figure 1.13: Probability of the outcomes of throwing the two dice in example 1.4.23

1.4.5 Law of total probability

A widely used proposition within probability theory is the so called *Law of Total Probability* given in equation 1.13.

Proposition 1.4.24. Consider a sequence $\{B_i\}$ such that $B_i \cap B_j = \emptyset$ for every $i \neq j$ and $B_1 \cup B_2 \cup \dots \cup B_n = \Omega$. That is, $\{B_i\}$ constitutes a so-called partition of a sample space. Then,

$$P(A) = \sum_{i=1}^n P(A|B_i) \cdot P(B_i) \quad (1.13)$$

Proof. Since $(A \cap B_i) \cap (A \cap B_j) = \emptyset$ for every $i \neq j$ (see Figure 1.14), then $A = \{(A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n)\}$

It follows that:

$$\begin{aligned} P(A) &= P\{(A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n)\} \\ &= P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n) \\ &= \sum_{i=1}^n P(A \cap B_i) = \sum_{i=1}^n P(A|B_i) \cdot P(B_i) \end{aligned} \quad (1.14)$$

The final step follows on the basis of the definition of conditional probability (1.4). $\square$

Example 1.4.25. A die is thrown twice and required is the probability that the sum of two outcomes is equal to 9. Denote by B_i the event that “the outcome of the first throw is i ” and by A the event that “the sum of the two outcomes is equal to 9”. The *law of total probability* (equation 1.13) is applied. If the first throw is a 1 or a 2, the probability that the sum of the outcomes in each throw is equal to 9 is 0. That is $P(A|B_1) = P(A|B_2) = 0$. If the first throw yields a 3, the second throw

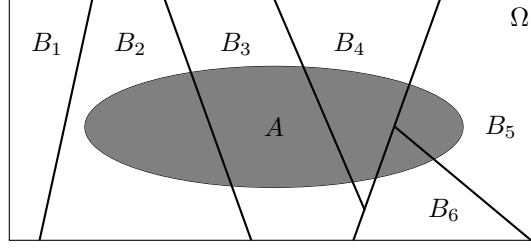


Figure 1.14: $A \cap B_i$ are mutually exclusive and their union is A

must be a 6, so $P(A|B_3) = 1/6 = P(B_3)$. The same reasoning applies with 4, 5 and 6, that is, $P(A|B_4) = P(A|B_5) = P(A|B_6) = 1/6$. Then,

$$\begin{aligned} P(A) &= \sum_{i=1}^6 P(A|B_i) \cdot P(B_i) = \sum_{i=3}^6 P(A|B_i) \cdot 1/6 \\ &= 1/6 \cdot \sum_{i=3}^6 P(A|B_i) = 4/36 \end{aligned}$$

It is of course also possible to calculate this result with other approaches. Consider for example table 1.3. These correspond to B_1 and B_2 . In events 18, 23, 28 and 33 the sum of the outcomes would end up in 9.

1.4.6 Bayes' Theorem

This famous theorem bears the name of reverend Thomas Bayes (1701(?) - 1761) who was credited by Richard Price as the first one to formulate the theorem (see Barnard and Bayes [1958]). Roughly, the theorem provides a technique for calculating a “posterior” distribution on the basis of a “prior”. With the same notation used in Section 1.4.5 we have:

$$\begin{aligned} P(A \cap B_i) &= P(A) \cdot P(B_i|A) \\ P(B_i \cap A) &= P(B_i) \cdot P(A|B_i) \end{aligned}$$

Which by equality implies that:

$$P(B_i|A) = \frac{P(A|B_i) \cdot P(B_i)}{P(A)} \quad (1.15)$$

This equation is known as Bayes' Theorem. It can be perceived as a rule for “information processing”. The probability of event B_i in the presence of event A is the “prior” probability of event B_i , that is $P(B_i)$, multiplied with the “likelihood” of A upon observing B_i (which is $P(A|B_i)$) and then normalized by $P(A)$.

Richard Price (see Barnard and Bayes [1958]) but also Laplace (Laplace et al. [1951]) used this theorem to explain the probability that the sun will rise every day. Laplace wrote: “Placing the most ancient epoch of history at five thousand years ago, or at 1826213 days, and the sun having risen constantly in the interval at each revolution of twenty-four hours, it is a bet of 1826214 to one that it will rise again tomorrow”. In other words, there is a probability of about 0.999999452419348 that the sun will rise the next day. We will come back to this problem in coming sections.

CHAPTER 2

Random Variables

In previous sections (and courses) we have already been dealing with random variables. What follows is a brief discussion of these for the discrete case. Intuitively speaking, random variables are those variables that are subject to variation which in turn may be described by a probability distribution function. In other words they are functions that take elements from a *sample space* (see section 1.3) and assign them a number in the interval $[0, 1]$. Roughly speaking, random variables may be discrete or continuous. In this section we will briefly discuss some examples of discrete models describing random variables.

2.1 Discrete random variables

These are variables that can take values on a finite outcome space. These have already been discussed in previous sections. Think for example on successive throws of a coin. The outcomes of throwing a coin may be only “heads” or “tails”. The outcomes of throwing a die may be $\{1, \dots, 6\}$ and so on.

Someone interested in traffic applications may think on counting the number of vehicles passing by a certain point in a particular period of time. Also, someone interested in cyber-security may be interested in the number of possible cyber-attacks in a particular interval of time. In Physics the number of electrons emitted by a radioactive source may be relevant. Notice that these last examples refer also to a discrete outcome which in principle may be very large. Possible outcomes of certain discrete random variables may even be infinite!

2.1.1 Probability density and cumulative distributions for discrete random variables

Discrete random variables may take particular values with positive probability. The function describing this probability is called probability mass function or probability density function (often referred to as *pdf* for short). We will denote it as:

$$f_X(x) = P(X = x)$$

The pdf of discrete random variables has many properties. Two of the most important ones are given next without a proof.

Proposition 2.1.1. *Let X be a discrete random variables then for all $x \in X$ it holds that $0 \leq f_X(x) \leq 1$ and $\sum_{x \in X} f_X(x) = 1$*

The pdf of a discrete random variable can be represented in different forms. The most common ones are through a function, a table or a graph. Discrete random variables occur naturally in applications. To see this consider Example 2.1.2 below.

Example 2.1.2. Consider the popular Weigh-in-Motion (WIM) systems that provides valuable data for structural, road, and transport engineers. WIM devices are designed to record the axle weights, gross vehicle weights, vehicle speed, inter-axial distance, vehicle type and other properties of heavy traffic as vehicles drive over a measurement site. WIM systems are capable of measuring vehicles traveling at a reduced or normal speed which makes them very attractive for different applications.

An example of the database for highway 12 in the Netherlands is presented in Table 2.1. For highway 12, the left direction indicates east to west. Only the slow lanes are considered since the data of interest corresponds to heavy traffic. The first column in Table 2.1 corresponds to vehicle's id, the second to the date and time of the registration of the data, the third to the type of vehicle and the fourth to the speed [km/h], the fifth to the road and direction, the sixth to vehicle length [m] and the seventh to vehicular weight [kN]. Finally one observation per axle load [kN] is contained within the database. Observe that vehicular weight is the sum of the individual axle load observations.

id	time	subcat	spd	highway	lngh	wght	a1	a2	a3	a4	a5	a6	a7	a8	a9	a10	a11	a12	a13
24515900	01/Apr/2013 00:06:20	T11O3	83	12L	16.6	425.5	66.7	104.0	85.0	84.4	85.3								
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
24516659	01/Apr/2013 06:45:35	T11O3	84	12L	15.6	147.3	52.1	41.6	17.3	18.2	18.2								
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
24518454	01/Apr/2013 11:04:28	V11	104	12L	7.2	42.8	18.3	24.5											
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
24525634	02/Apr/2013 03:27:26	V1111A12	85	12L	20.5	396.3	43.3	48.7	35.9	39.0	82.8	75.5	71.1						
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
24746989	05/Apr/2013 14:00:57	O	84	12L	28.9	631.9	78.3	72.8	65.4	64.1	48.6	48.9	51.6	52.3	49.0	47.8	52.9		
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
26436033	01/May/2013 00:00:06	T11O3	81	12L	16.6	216.2	66.8	61.5	29.6	29.4	28.9								

Table 2.1: Example of WIM database highway 12L in the Netherlands.

Consider the random variable X , the number of axles per vehicle registered by the WIM system in April 2013. The pdf of this random variable is represented graphically in Figure 2.1. Figure 2.1 is computed with the data in Table 2.2 normalized by the 165,281 observations (vehicles) in the month of April 2013 on highway 12L in the Netherlands. Notice that this variable is not explicitly observed in Table 2.1 however it may be inferred from the data as is presented.

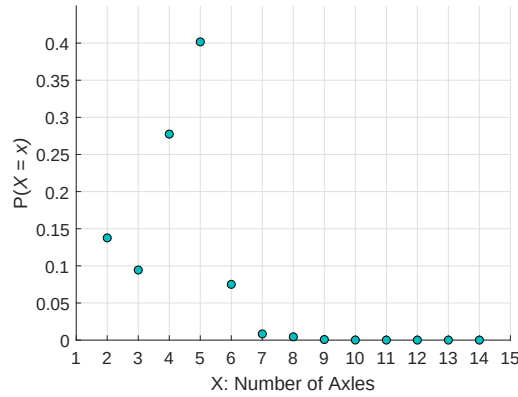


Figure 2.1: Probability density function for the number of axles of vehicles observed in the WIM data represented in Table 2.1

Number of Axles	2	3	4	5	6	7	8
Number of Vehicles	22,750	15,605	45,855	66,383	12,413	1,376	733
Number of Axles	9	10	11	12	13	14	
Number of Vehicles	128	20	10	4	3	1	

Table 2.2: Number of vehicles according to their number of axles.

The cumulative distribution function is the function that describes the probability that a random variable takes values less than or equal to a particular value. The cumulative distribution function is often referred to as *cdf*. This function is given by equation 2.1.

$$F_X(x) = P(X \leq x) = \sum_{x \in \{X \leq x\}} f_X(x) \quad (2.1)$$

Example 2.1.3. The cdf corresponding to the random variable whose pdf is represented in Figure 2.1 is presented in Figure 2.2.

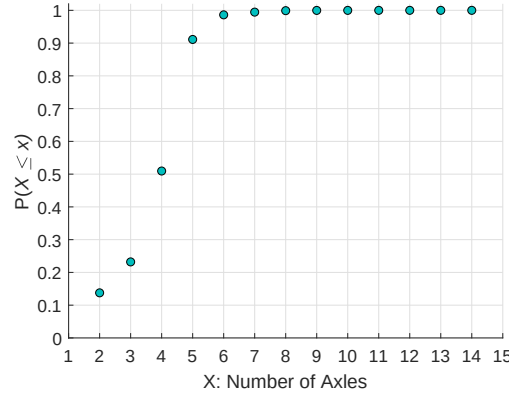


Figure 2.2: Cumulative function for the number of axles of vehicles observed in the WIM data represented in Table 2.1

Notice that other probabilities of interest may be computed from the pdf or cdf. For example the probability that X lies between two possible outcomes x_1 and x_2 is:

$$P(x_1 \leq X \leq x_2) = \sum_{x \in \{x_1 \leq X \leq x_2\}} f_X(x)$$

Or the probability that X is larger than a particular value x is given in Equation 2.2

$$P(X > x) = 1 - F_X(x) \quad (2.2)$$

2.1.1.1 Expected value and Variance for Discrete Random Variables

Mathematical expectation (or expectation or expected value) is an important measure of central tendency for random variables. This measure is sometimes also referred to as the mean value of a random variable. The definition of expectation (when finite), keeping the notation as above, is given by equation 2.3.

$$E(X) = \sum_{x \in X} x \cdot f_X(x) \quad (2.3)$$

Suppose that $P(X = x) = 1/r$ for all $x \in X$, then it is easy to see that $E(X) = \frac{x_1 + x_2 + \dots + x_r}{r}$. In this case $E(X)$ is the arithmetic average of all $x \in X$. Notice that equation 2.3 indicates that $E(X)$ is a weighted average of the values of $x \in X$, where the weights are given by their individual probabilities $f_X(x)$. Mathematical expectation has interesting properties that make it also useful as an operator. Some of these properties are given next without proof.

Proposition 2.1.4. *Let X and Y be two discrete random variables with finite expectation and denote by $g(\cdot)$ some real-valued function of the random variable X and let a and b denote two constants. Then:*

$$E(g(X)) = \sum_{x \in X} g(x) \cdot f_X(x)$$

$$E(a) = a$$

$$E(aX + bY) = aE(X) + bE(Y)$$

Suppose $P(X \leq Y) = 1$ then $E(X) \leq E(Y)$ with strict equality if and only if $P(X = Y) = 1$

$$|E(X)| \leq E(|X|)$$

Another important summary measure for discrete random variables is the *variance*. The variance is a measure of dispersion around the mean. The variance (just as mathematical expectation) is based on *moments*. Let X be a discrete random variable and let $r \geq 0$ be an integer. We say that X has a moment of order r if X^r has finite expectation and in that case $E(X^r)$ is defined as the r -th moment of X . Notice that the variance is the expectation of the squared deviations of a random variable to its mean. Alternatively, the variance is the difference between the second moment and the squared of the first moment.

$$\text{Var}(X) = E[(X - E(X))^2] = E(X^2) - E(X)^2 \quad (2.4)$$

The variance of X is often denoted as σ_X^2 or $\sigma^2(X)$. The standard deviation is the squared root of the variance (σ_X or $\sigma(X)$). Here we will use both notations for the variance and standard deviation.

The variance also has important properties (like expectation) that makes it attractive as mathematical operator. We will come back to these later.

2.2 Common parametric discrete distributions

2.2.1 The binomial distribution

Consider an experiment which consist of n *identical* and *independent* trials with two possible outcomes in each trial. The probability of success (let the number 1 denote success) in one trial is p and is constant across trials. Each trial is independent and the variable of interest is X , the number of successful outcomes in n trials. The random variable X follows a binomial distribution. The density function of a binomial random variable is:

$$f_X(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad (2.5)$$

An example of such variable is tossing a coin (perhaps not a fair coin) where the probability of “success” (observing heads) is given by p . A possible sequence of 5 tosses is 1, 1, 1, 1, 1 that is, all outcomes are a success. Another one is 0, 1, 1, 0, 1 that is three successes in 5 tosses. Notice that in general there are $\binom{n}{x}$ possible sequences of x successes in n trials all with probability p . Notice that for each of the $\binom{n}{x}$ sequences with x successes there will be $n - x$ outcomes that are not successful each with probability $(1 - p)$. Hence, the derivation of equation 2.5. In Figure 2.3 presents an example of the density function of a binomial random variable with 10 expected successes in 20 trials. Figure 2.4 presents the cumulative distribution function of the same binomial random variable.

The mean and variance of a binomial random variable X are given by:

$$E(X) = np \quad (2.6)$$

$$\text{Var}(X) = np(1 - p) \quad (2.7)$$

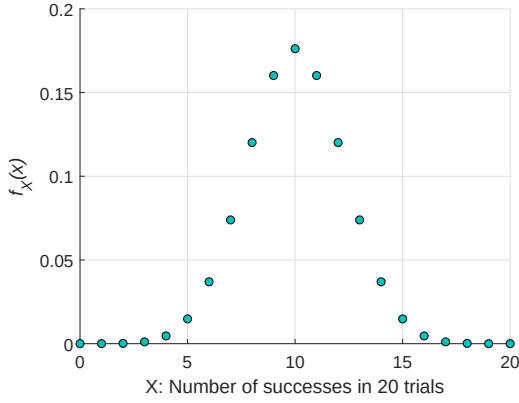


Figure 2.3: Binomial density function.

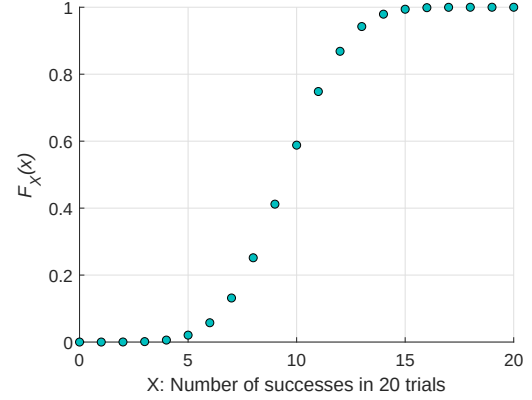


Figure 2.4: Binomial cumulative distribution function.

2.2.2 The geometric distribution

Consider a similar experiment as described in section 2.2.1. The variable of interest in this case is however X , the number of trial until one would observe the first success. This random variable follows a geometric distribution. The density function of a geometric random variable is:

$$f_X(x) = P(X = x) = p(1 - p)^{x-1} \quad (2.8)$$

The derivation of the density function in (2.8) follows the same reasoning as that used to derive the binomial density. There will be one success in the x^{th} trial with probability p and there will be $x - 1$. Examples of the density and cumulative distribution functions for a geometric random variable are presented in figures 2.5 and 2.6 respectively.

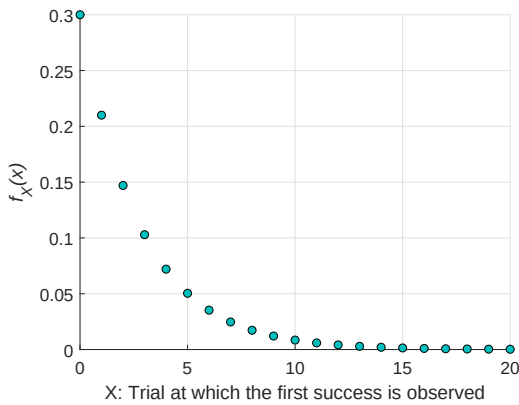


Figure 2.5: Geometric density function.

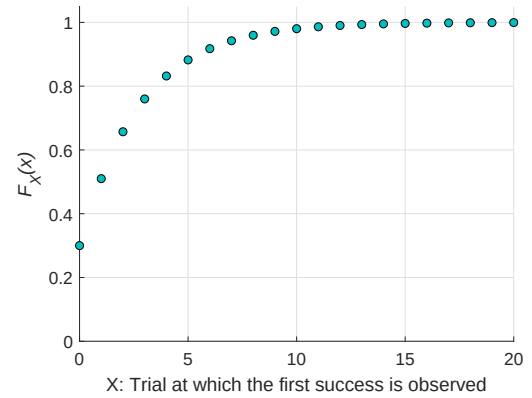


Figure 2.6: Geometric cumulative distribution function.

The mean and variance of a geometric random variable X are given by equations (2.9) and (2.10).

$$E(X) = \frac{1}{p} \quad (2.9)$$

$$\text{Var}(X) = \frac{1-p}{p^2} \quad (2.10)$$

2.2.3 The negative binomial distribution

Consider again independent trials each of which may have a success with probability p and a failure with probability $(1 - p)$. The random variable of interest X is in this case the number of trials at which we observe the r^{th} success. X will follow a negative binomial distribution. In order to derive the density function consider first the event A : the first $(x - 1)$ trials contain $(r - 1)$ successes. For $x \geq r$, A follows a binomial distribution and hence according to equation (2.5) we have $P(A) = \binom{x-1}{r-1} p^{r-1} (1-p)^{x-r}$. The probability of observing a success on the x^{th} trial is p and hence the density function of X is given by:

$$f_X(x) = P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r} \quad (2.11)$$

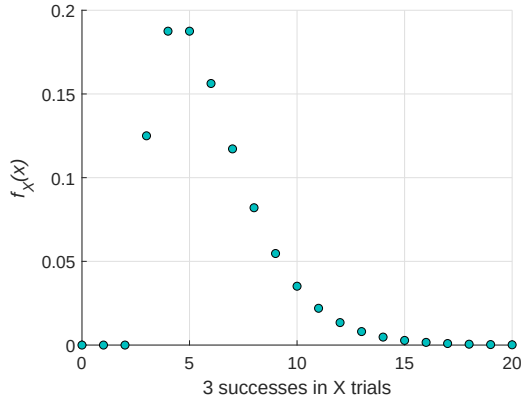


Figure 2.7: Negative binomial density function.

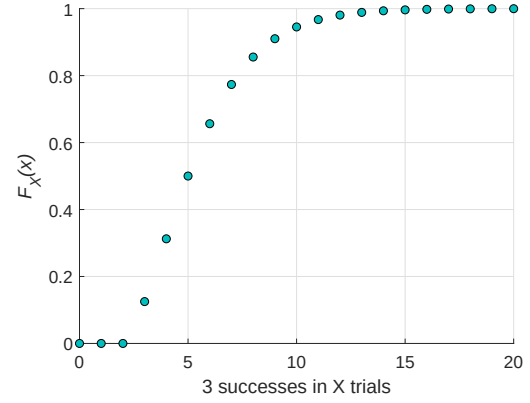


Figure 2.8: Negative binomial cumulative distribution function.

The mean and variance of a negative binomial random variable X are given by equations (2.12) and (2.10).

$$E(X) = \frac{r}{p} \quad (2.12)$$

$$\text{Var}(X) = \frac{r(1-p)}{p^2} \quad (2.13)$$

CHAPTER 3

System Reliability

3.1 Fault trees.

Fault Tree analysis is considered a technique which allows the analysis of a system in the context of its environment and operation to find the largest number of credible ways in which an undesired state of a system may happen [Vesely et al., 1981, p.IV-1]. Consider for example a system that consists of three components. The Failure of the system will be denoted as A_4 . The failure of each component will be denoted as A_1, A_2 and A_3 . From the characteristics of the system we know that it fails whenever A_1 happens simultaneously with A_2 **OR** A_3 . Two graphical representations of the event of system failure are presented in figures 3.1 and 3.2.

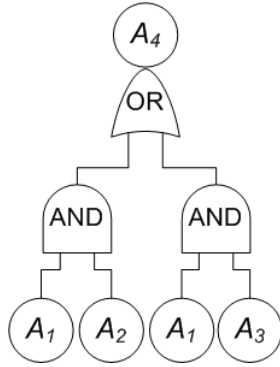


Figure 3.1: System failure represented as a fault tree.

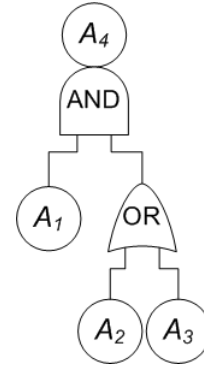


Figure 3.2: Equivalent system failure represented as a FT.

Basic events in Fault Trees can be represented by a Boolean variable. In this sense, a FT may be thought of as a picture of a Boolean formula. In Boolean arithmetic, variables take only two values, usually 0 or 1.

Suppose A_1 and A_2 are Boolean variables, then $A_1 \oplus A_2$ and $A_1 \otimes A_2$ are also Boolean variables, and hence take values 0 or 1. This is arranged by defining:

- $A_1 \oplus A_2 = A_1 + A_2 - A_1 \cdot A_2$ and
- $A_1 \otimes A_2 = A_1 \cdot A_2$.

The operators $\otimes$ and $\oplus$ correspond to the **AND** and **OR** operators in propositional logic. In other words, $A_1 \oplus A_2$ means “either A_1 or A_2 or both are true” and $A_1 \otimes A_2$ means “both A_1 and

A_2 are true”. In Boolean arithmetic $A_1 \otimes A_1 = A_1$. This corresponds to saying that the event A_1 AND A_1 is the same as the event A_1 .

Other rules for Boolean algebra may be found in [Vesely et al., 1981, p.VII-2]. The rules of boolean algebra are the mathematical building block of Fault Tree analysis. The reader may refer to [Vesely et al., 1981, p.VII-2] for an introductory treatment of boolean algebra in FT analysis. In the two examples presented in Figures 3.1 and 3.2 the usual notation for AND and OR gates is displayed.

In most cases, we do not know whether a given basic event occurs, we know only its probability of occurrence. Recall that $E(A_1) = P(A_1 = 1)$ where E denotes mathematical expectation. If A_1 and A_2 are independent, then:

- $E(A_1 \oplus A_2) = E(A_1) + E(A_2) - E(A_1) \cdot E(A_2)$ and
- $E(A_1 \otimes A_2) = E(A_1) \cdot E(A_2)$

The above reasoning might suggest that we can just replace the Boolean variables at the base of a fault tree with their probabilities (i.e. their expectations) and compute the probability of the top event with ordinary arithmetic. This is not true, in general, and it may depend on how the fault tree is displayed. This is precisely illustrated in Figures 3.1 and 3.2.

If Boolean reduction is applied to both FTs we see that they yield the same formula for A_4 . However, if we replace the variables by their expectations and apply ordinary arithmetic to the non-reduced formulae, we will get different answers. The correct calculation would be obtained from Figure 3.2:

- $P(A_4 = 1) = P(A_1 = 1) \cdot P(A_2 = 1) + P(A_1 = 1) \cdot P(A_3 = 1) - P(A_1 = 1) \cdot P(A_2 = 1) \cdot P(A_3 = 1)$.

The problem is that when $(A_1 \text{ AND } A_2) \text{ OR } (A_1 \text{ AND } A_3)$ is computed with expectations in Figure 3.1, the term $P(A_1 = 1)$ is included twice.

In general, computing probabilities of occurrence from a Fault Tree requires some careful manipulations, before substituting Boolean variables with their expected values. However, if our Fault Trees contain no “repeated events” then we can replace Boolean variables with expectations and replace Boolean arithmetic with ordinary arithmetic. This assumes that we have captured all common cause dependencies in the Fault Tree. This means that once probabilities are assigned to the basic events, the probability of joint occurrence is computed as the product of the probabilities.

Fault trees provide with a static (in time) representation of the reliability of the system. By taking advantage of their graphical notation they also provide with a good description of the logic of the system. FTs have been widely used in reliability analysis.

Example 3.1.1. Consider a solid state light system that consists of three LEDs. The system may fail when one of the three LEDs fail catastrophically or they depreciate such that only a certain percentage of the total luminance is achieved. Further more, the system may fail due to failure of the solder of each led or due to a driver failure. The graphical representation of the FT for this system is presented in figure 3.3.

The LED failure may be due to any combination of each of the three individual LEDs failing catastrophically taken one, two or three at a time. It may also be due to depreciation of the three LEDs. However the **XOR** gate indicates that both types of failure may not be observed at the same time. This makes the events of catastrophic and depreciation failure dependent and this dependence needs to be considered in the calculus of the total system reliability. $\square$

3.2 Bayesian Networks

We will understand *Bayesian Networks* as Directed Acyclic Graphs (DAGs) whose nodes represent univariate random variables and whose arcs represent direct qualitative dependence relationships between variables represented by adjacent nodes. The direct predecessors (successors) of a node are called parents (children). A DAG is presented in figure 3.4. In terms of BNs figure 3.4 states that the values of variable X_2 depend on the values of variable X_1 . Similarly the values of X_3 depend on X_2

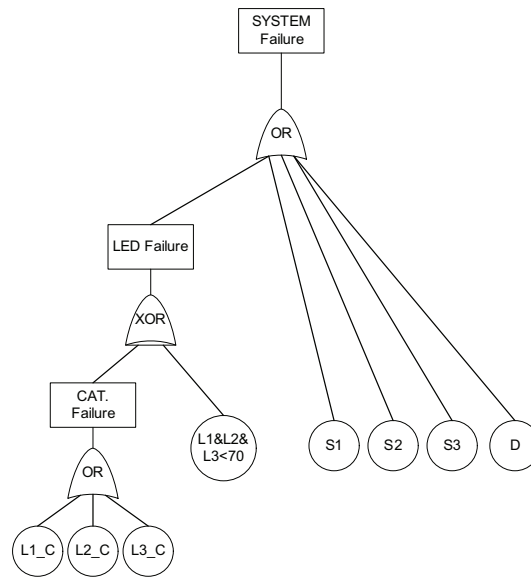


Figure 3.3: Fault Tree for three LEDs system.

and X_1 . An example of a graph that is not a BN is presented in figure 3.5. Intuitively, one may see that a directed graph with cycles does not represent a BN because X_1 would require ‘information’ from itself (that is through the path $X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_1$) to explain itself.

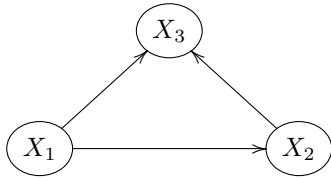


Figure 3.4: Bayesian Network on three nodes.

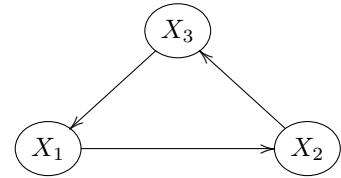


Figure 3.5: Directed graph with a cycle.

The absence of arcs guarantees a set of (conditional) independence facts. An univariate marginal/unconditional distribution is specified for each node with no parents (X_1 in figure 3.4), and a conditional distribution is associated with each child node. The (conditional) distributions serve as the quantitative information about the strength of the dependencies between the variables involved.

The graph with the conditional independence statements encoded by it, together with the (conditional) distributions represents the joint distribution over the random variables denoted by the nodes of the graph. The quantitative information needed can be retrieved from data. When data is not available, structured expert judgment could be used to quantify the random variables themselves and their interdependence in the context of a given model.

The relatively simple visualization of the complicated relationships among the random variables is one of the most appealing features of a BN model. The main use of BNs is to update distributions given observations. If some variables have been observed, we want to infer the probabilities of others which have not yet been observed. This is referred to as inference in BNs.

The influences represented by the arcs of a BN may be probabilistic or deterministic. When an influence is deterministic, nodes will be called *functional*. The graph of a BN induces a non unique ordering of variables and stipulates that each variable is conditionally independent of its

non-descendants given its parents. The parent set of variable X_i will be denoted as $Pa(i)$. Hence, to specify a joint distribution through a BN the graph must be specified together with conditional probability functions of each variable given its parents (equation (3.1)).

$$f(X_1, \dots, X_n) = \prod_{i=1}^n f_{X_i|X_{Pa(i)}} \quad (3.1)$$

If $Pa(i) = \emptyset$ then $f_{X_i|X_{Pa(i)}} = f(X_i)$. A BN is then a concise and complete representation of the joint distribution. We start by presenting discrete BNs.

3.2.1 Discrete Bayesian Networks

In the case that all nodes in the BN are discrete then the functions to be specified are conditional probability tables (CPT) of each node given its parents. For example, suppose that all variables in the BN on figure 3.4 were discrete with states ‘failed’ and ‘not failed’. Then, in order to quantify the BN first for X_1 we would require to specify $P(X_1 = \text{failed})$ or $P(X_1 = \text{not failed})$. Only one of the two probabilities is required because the remaining may be computed from the relation $P(X_1 = \text{failed}) = 1 - P(X_1 = \text{not failed})$.

For X_2 the table in 3.1 would be required. Notice that table 3.1 requires four entries, however two may be computed by difference since all conditional probabilities must sum to one. For X_3 table 3.2 would be required. Similarly to the case of X_2 , 3 (instead of 6) conditional probabilities would need to be specified. The rest may be again determined by difference.

	$P(X_2=\text{failed} X_1 = x_1)$	$P(X_2= \text{not failed} X_1 = x_1)$
$x_1=\text{failed}$		
$x_1=\text{not failed}$		

Table 3.1: Conditional probability table for X_2 .

	$P(X_2=\text{failed} X_1 = x_1, X_2 = x_2)$	$P(X_2= \text{not failed} X_1 = x_1, X_2 = x_2)$
$x_1=\text{failed}, x_2= \text{failed}$		
$x_1=\text{failed}, x_2= \text{not failed}$		
$x_1=\text{not failed}, x_2= \text{failed}$		
$x_1=\text{not failed}, x_2= \text{not failed}$		

Table 3.2: Conditional probability table for X_2 .

3.2.1.1 Quantification

In general the number of probabilities to be assessed for a discrete BN on n nodes with k_i states for each X_i for $i = 1, \dots, n$ is:

$$\kappa = \sum_{j \in S} k_j - |S| + \sum_{l \in C} \left((k_l - 1) \prod_{m \in M(l)} k_m \right) \quad (3.2)$$

where $S = \{j : Pa(X_j) = \emptyset\}$, $C = \{l : Pa(X_l) \neq \emptyset\}$ with $|S| + |C| = n$ and $M(l) = \{m : X_m \in Pa(X_l)\}$. In the absence of data conditional probabilities required need to be assessed by experts. Because κ grows very fast with the number of nodes and the number of states of each node,

unfortunately, in large, complex structures this leads to huge conditional probability tables that have to be quantified in a defensible way. Moreover, even when the quantification could be done, exact inference might prove unfeasible due to the large number of calculations required to perform it Edwards [1998]. Fault trees may often be represented as Bayesian Networks.

Example 3.2.1. Consider the system from example 3.1.1. A discrete BN representing the system is presented in figure 3.6. Probabilities for variables LED1.CAT, LED2.CAT, LED3.CAT, Solder_1, Solder_2, Solder_3 and Driver need to be assessed. All values used in figure 3.6 are fictitious and for illustration purposes only.

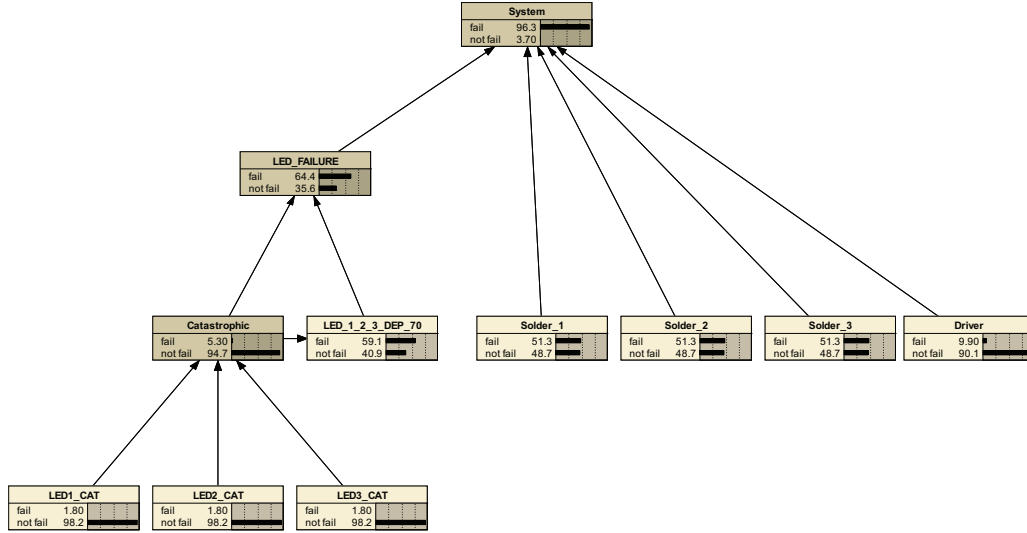


Figure 3.6: Bayesian Network representing the Fault Tree for three LEDs system in figure 3.3.

For the quantification of the catastrophic failure, observe that a led failure will be observed when at least one LED fails catastrophically. Thus:

$$P(\text{Catastrophic} = \text{not fail} | \text{LED1.CAT} = \text{not fail}, \text{LED2.CAT} = \text{not fail}, \text{LED3.CAT} = \text{not fail}) = 1 \quad (3.3)$$

If at least one of the conditioning variables observe the state *fail* then the conditional probability in equation 3.3 would change to 0. The **XOR** gate in figure 3.3 makes the events of catastrophic and depreciation failure dependent. This is observed in the BN from figure 3.6 by drawing an arc from variable Catastrophic to variable LED_1.2.3_DEP_70. In this case we consider the conditional probabilities in relation 3.4 with their complements.

$$\begin{aligned} P(\text{LED_1_2_3_DEP_70} = \text{fail} | \text{Catastrophic} = \text{fail}) &= 0 \\ P(\text{LED_1_2_3_DEP_70} = \text{fail} | \text{Catastrophic} = \text{not fail}) &= 0.6248 \end{aligned} \quad (3.4)$$

Variables LED.failure and System are quantified similarly as the **OR** gate for Catastrophic failure (see equation 3.3). □

3.2.1.2 Inference

The main use of BNs is to update distributions given observations. This is referred to as inference in BNs. Inference in discrete BNs can be performed using either exact algorithms (e.g. Zhang and Poole [1994], Pearl [1988]), or approximate algorithms (e.g. Jordan et al. [1999], Murphy [2002]).

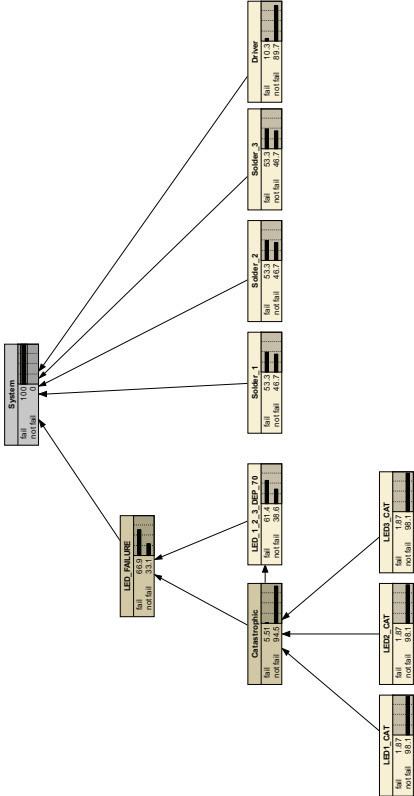


Figure 3.7: Evidence concerning no catastrophic failure.

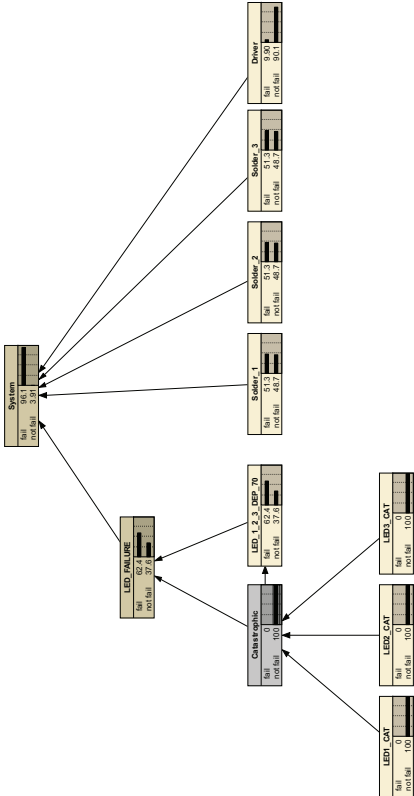


Figure 3.8: Evidence concerning system failure.

Example 3.2.2. Figure 3.7 presents the updated joint distribution of the variables in figure 3.6 given the knowledge that no catastrophic failure has occurred. Observe that for example depreciation failure becomes more likely and equal to LED failure.

Figure 3.7 presents the updated joint distribution of the variables in figure 3.6 given the knowledge that no system failure has occurred. According to the model, all possible causes of failure have increased in probability. $\square$

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